Delhi Public School
Gurugram | Sector 67A
Nurturing the Whole Child

WINTER HOLIDAY ASSIGNMENT
SESSION: 2025-26
GRADE: IX
SCIENCE

General Instructions:

- Solve the following questions in your science notebook.
- There are three sections -Section A Biology, Section B -Chemistry, Section C Physics.
- There are 7 pages in all.
- Attempt 10 questions each day.
- Write the answers neatly.
- Submission Date 8.01.25

SECTION A-BIOLOGY

OBJECTIVE TYPE QUESTIONS :

1. Cell theory states that all organisms are made up of one or more similar units of organization called cells.
Which of the following organisms do not strictly adhere to this theory?
a) Protozoa b) Bacteria c) Viruses d) Algae

2. A plant cell placed in a hypotonic solution will not burst because of presence of :

a) Plasma membrane b) Cell wall c) Chloroplast d) Cytoplasm

3. Plant cell wall is mainly composed of:

a) Sugars b) Cellulose c) Proteins d) Lipids

4. A cell will swell up if:

a) The concentration of water molecules in the cell is higher than the concentration of water molecules in the surrounding medium.
b) The concentration of water molecules in the surrounding medium is higher than water molecules concentration in the cell.

c) The concentration of water molecules is same in the cell and in the surrounding medium
d) Concentration of water molecules does not matter.

5.What is a basis for differentiation of a prokaryotic cell from a eukaryotic cell?

a) Presence or absence of cytoplasm
b) Presence or absence of cell membrane
c) Presence or absence of genetic material
d) Presence or absence of nuclear membrane.

6. Which of the following would you not find in a bacterial cell?

a) DNA b) Cell membrane c) Golgi apparatus d) Ribosomes

7. The image shows cells in the onion peel and human cheek. What can be understood by observing these cells?

a) All living things are made up of cells that look similar.
b) All living things are made up of cells that are structurally similar but functionally different.
c) All living things are made up of cells that look different from each other
d) None of the above.

8. The main function of a plasma membrane is to:
a) Prevent water from entering or leaving.
b) Control what goes into and out of the cell.
c) Act as a sieve, allowing only lipids to pass.
d) Move the cell from place to place.

ASSERTION AND REASON

For the questions 13 to 16, two statements are given-one labelled **Assertion(A) and the other labelled Reason(R).** Select the correct answer to these questions from the options (i) , (ii) ,(iii) and (iv)as given below:

(i) Both A and R are true and R is the correct explanation of the assertion.
(ii) Both A and R are true but R is not the correct explanation of the assertion.
(iii) A is true but R is false.
(iv) A is false but R is true.

8.Assertion (A): The shape of the cells is of different types ranging from circular, elongated, tubular, oval, cylindrical, etc.
Reason(R): The shape of the cells varies according to the specific function they perform.

9. Assertion (A): Cell wall is found in plant cell.
Reason(R): Animal cells are covered only by cell membrane.

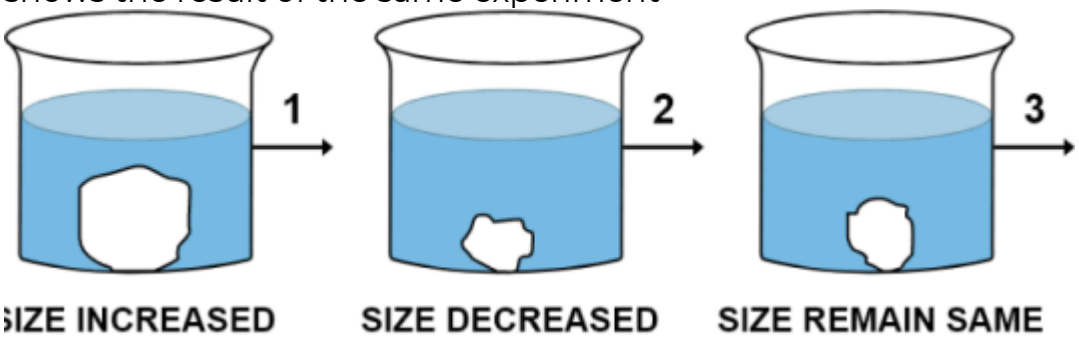
10. Assertion (A): Cell wall is a non-living part of the cell.
Reason(R): It does not offer protection, definite shape and support.

Long Answer questions
11. Explain in detail what do you know about the structure of nucleus.

12 Draw a neat labelled diagram of a prokaryotic cell

13. Describe the process of diffusion of O_2 and CO_2 through the cell membranes.

14.A candidate in order to study the process of osmosis has taken 3 potato cubes and put them in 3 different beakers containing 3 different solutions. After 24 hours, in the first beaker the potato cube increased in size, in the second beaker the potato cube decreased in size and in the third beaker, there was no change in the size of the potato cube. The following diagram shows the result of the same experiment



a) Give the technical terms of the solutions used in the beakers 1, 2, and 3. In beaker 3, the size of the potato cube remains the same. Explain the reason in brief.
b) What is osmosis?
c) How do a cell wall and a cell membrane differ in their permeability?

15. State the function of blood.

16.What is ciliated columnar epithelium? What is the function of ciliated columnar epithelium?

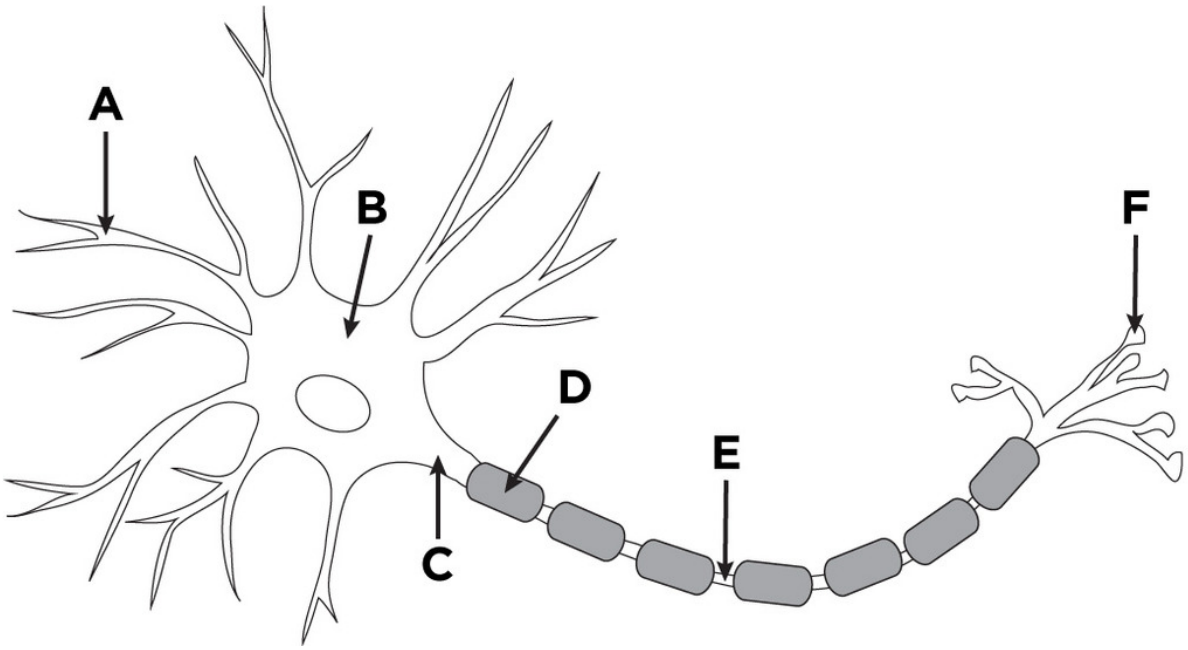
17. Differentiate between voluntary and involuntary muscles. Give one example of each type.

18. (i) What is the lining of blood vessels made up?
(ii) What is the lining of small intestine made up of?

19. Why do animals in the arctic region have thicker layer of fat-storing tissue below their skin?

20. Draw well labelled diagrams of various types of muscles found in human body

21. Label the given diagram and explain how messages a retransferred in neuron.



SECTION- B(PHYSICS)

1 List two conditions which need to be satisfied for the work to be done on an object.

2 A coolie is walking on a railway platform with a load of 27 kg on his head. What is the amount of work done by him?

3 a. Give one example each of potential energy
(i) due to position
(ii) due to shape.

(b) Derive an expression for kinetic energy of an object.
(c) If the velocity of an object is doubled. What will be change in its kinetic energy?

4.A rocket is moving up with a velocity v. If the velocity of this rocket is suddenly tripled, what will be the ratio of two kinetic energies?

5. A light and a heavy object have the same momentum, find out the ratio of their kinetic energies.
Which one has a larger kinetic energy?

6.Numericals :-

i. The time period of a vibrating object is 0.01 s.
Find the frequency of the vibration.

ii. The speed of sound in air is 340 m/s.
Calculate the wavelength of a sound wave whose frequency is 170 Hz.

iii. An echo is heard after 2 seconds.
If the speed of sound is 340 m/s, calculate the distance of the reflecting surface from the source.

iv. An echo is heard after 2 seconds.
If the speed of sound is 340 m/s, calculate the distance of the reflecting surface from the source.

v. A tuning fork produces 600 vibrations in 3 seconds.
Calculate the frequency of the tuning fork.

vi. A person standing near a cliff hears an echo after 3 seconds.
If the speed of sound in air is 340 m/s, calculate the distance between the person and the cliff.

vii. An ultrasound wave has a frequency of 1 MHz and travels through human tissue at a speed of 1540 m/s.
Calculate the wavelength of the ultrasound wave.

viii. A force of 20 N acts on a body and moves it through a distance of 5 m in the direction of the force.
Calculate the work done by the force.

ix. A stone of mass 2 kg is raised to a height of 5 m.
Calculate its gravitational potential energy.

x. A motor lifts a load of 80 kg to a height of 10 m in 5 seconds.
Calculate:
a) Work done
b) Power of the motor

xi. A stone of mass 1 kg is thrown vertically upward with a speed of 20 m/s.
Calculate the height at which its kinetic energy becomes half of its initial value.

xii. Two machines A and B lift the same load of 200 kg to a height of 10 m.
Machine A takes 20 s and Machine B takes 10 s.
Calculate the power of each machine and compare them.

xiii. A lift carries a load of 500 kg upward through a height of 15 m in 10 s.
Calculate the power developed by the lift.

SECTION- C (CHEMISTRY)

1 In what way is the Rutherford's atomic model different from that of Thomson's atomic model?

2 Give two important applications of radioactive isotopes.

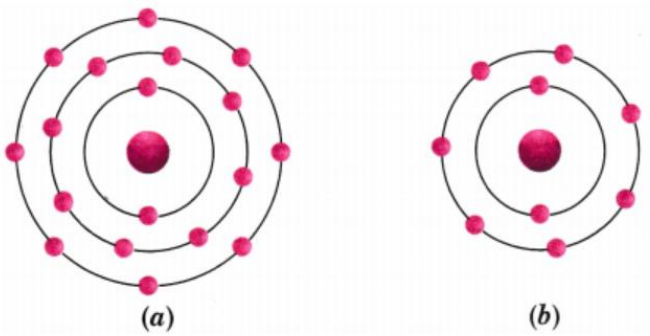
3 Predict the valency of the following elements
(i) A (Atomic number 5)
(ii) B (Atomic number 12)
(iii) C (Atomic number 14)
(iv) D (Atomic number 17)

4 The atomic number of lithium is 3. Its mass number is 7.
(a) How many protons and neutrons are present in a lithium atom?
(b) Draw the diagram of a lithium atom.

5.What is the gold foil experiment? Name the scientist who performed this experiment. Write the conclusions and shortcomings of Rutherford's model of atom.

6. Numericals:-

i. An element 'A' has valency +3, while another element 'B' has valency -2. Give the formula of their compound formed when 'A' reacts with 'B'.
ii. Write the electronic configuration of a positively charged sodium ion (Na^+). Atomic number of sodium is 11.
iii. The electronic configuration of phosphorus atom is 2, 8, 5. Give the electronic configuration of P^{3-} ion.
iv. The atomic number of Al and Cl are 13 and 17, respectively. What will be the number of electrons in Al^{3+} and Cl^-?
v. Write down the electron distribution of chlorine atom. How many electrons are there in the L shell? (Atomic number of chlorine is 17)
vi. Find out the valency of the atoms represented by the Figs. (a) and (b)



vii. The atomic number of lithium is 3. Its mass number is 7.
(a) How many protons and neutrons are present in a lithium atom?
(b) Draw the diagram of a lithium atom.

viii. Write the molecular formulae for the following compounds:
(a) Copper (II) bromide
(b) Aluminium (III) nitrate
(c) Calcium (II) phosphate
(d) Iron (III) sulphide
(e) Mercury (II) chloride
(f) Magnesium (II) acetate

ix. An atom 'M' of an element reacts with oxygen to form M_2O_3. Calculate the valency of the element 'M'.